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ABSTRACT

The effects of different high-altitudes on the blast parameters—including peak overpressure and specific impulse—produced by a center-initiated cylindrical charge (CC) were simulated using the AUTODYN software and analyzed through dimensional analysis theory. The relational equations of the blast parameters under reduced temperature and pressure were developed and validated through numerical simulations. The results demonstrate that the equations are effective in assessing the blast parameters generated by the center-initiated CCs under the conditions of reduced temperature, diminished pressure, and high-altitude environments effectively. The peak overpressure and impulse reduced on average by 29.6% and 40.0%, respectively, while the blast wave action range increased by 34.2% as the altitude changed from 0 to 10 000 m. When isolating the effects of diminished temperature and low pressure, it is found that low temperature reduced overpressure (impulse) by ~0.5% (15.2%) and decreased the action range by 17.2%. In contrast, diminished pressure led to an average reduction of 30.0% and 48.4% in overpressure and impulse, respectively, while increasing the action range by 58.9%. In high-altitude settings, the blast parameters and action range of the blast wave generated by center-initiated CCs are predominantly influenced by reduced pressure, whereas the impact of low temperature is marginal comparatively.

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I. INTRODUCTION

In harsh environments, such as high plateaus, the atmospheric pressure and temperature are markedly reduced compared to sea-level regions. The dynamic interplay between diminished pressure and low temperature in high-altitude environments exerts a profound influence on overpressure and impulse, which are vital parameters determining the intensity of explosive loads on structures.[1] Current guidelines, including UFC 3-340-02[2] and ASCE/SEI 59-11,[3] are commonly employed to assess blast loads. These guidelines assume that the explosive charges are spherical and the surrounding atmosphere is at normal temperature and pressure (NTP), defined as p_0 = 101.325 kPa and T_0 = 15 °C. However, in both military and civilian applications, most of the charges are designed to be cylindrical rather than spherical in geometry.[4] In addition, such explosive events are often conducted in high-altitude areas, where environmental conditions come into play during explosions and can influence the blast effects.[5] It is imperative to evaluate the physical manifestations of blast waves resulting from cylindrical charges (CCs) across various high-altitude settings. This assessment holds significance in enhancing the effectiveness of charges, advancing blast-resistant design, and mitigating unwanted consequences in plateau areas.

Considerable studies have focused on examining the blast characteristics produced by CCs under NTP ambient conditions. Ismail and Murray,[6] for instance, conducted a series of experiments involving CCs for free air bursts in NTP environments. Their findings revealed that the blast wave patterns were highly intricate, with multiple shocks appearing in the pressure-time histories. Knock and Davies[7,8] conducted experimental studies to investigate the effects of the charge length-to-diameter (L/D) ratio on peak overpressure and impulse under NTP conditions. In addition, they developed

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empirical formulas to describe the relevant parameters. Wu *et al.*[9] researched, both numerically and experimentally, the distributions of blast loads from bare SCs and CCs in NTP environmental conditions, detecting that the blast loads generated from CC are different from SC in the near field. Recently, Xiao *et al.*[10] and Gao *et al.*[11] carried out numerical analyses on the distribution of blast loads for CCs in NTP conditions. In summary, current studies on the characteristics of blast waves generated from CCs have predominantly been conducted in NTP ambient conditions.

Currently, several studies have examined the blast characteristics of spherical charges (SC) in low-temperature and pressure (LTP) environments. Izadifard and Foroutan,[12] for example, conducted a numerical study on SC blast characteristics in low-pressure environments, proposing modified models to describe the blast parameters under such conditions. Veldman *et al.*[13] conducted a comprehensive investigation, employing both numerical and experimental methods, to study blast loads generated by bare C-4 SCs under different atmospheric pressure conditions. The research revealed that a reduction in ambient pressure led to a substantial decrease in the conventional blast parameters. Wang and Kong[14] conducted measurements of the SC blast wave at various altitudes and analyzed the attenuation effects of altitude differences on the blast wave propagation. Silnikov *et al.*[15] analyzed the overpressure and impulse of SC blast waves under both normal and reduced pressure conditions and found that the blast effect was less pronounced in the case of an explosion taking place at reduced ambient pressure. Chen *et al.*[16] examined the propagation of blast waves generated from SC under LTP conditions, finding that the ambient environments greatly influenced the blast characteristics. To date, only a few studies have specifically addressed the influence of environmental pressure and temperature on blast loads, primarily assuming that the charges are spherical in shape.

CCs play a crucial role in both military and civil applications, and the diminished pressure and low temperature of high-altitude environments have a substantial impact on blast loads. Therefore, this paper examines the impact of high-altitude environments, along with the decoupling effects of LTP conditions, on the peak overpressure, specific impulse, and blast wave action range generated by center-initiated CCs. The results offer a reference for the investigations of the CCs blast characteristics in high-altitude regions and LTP conditions.

II. NUMERICAL MODELS

A. Development of the numerical model

A free-air burst using 1 kg of TNT with a density of ρ = 1.63 g/cm^3 is conducted to examine how charge shape influences blast parameters, including peak overpressure (Δp_m) and specific impulse (i), at different high-altitude and LTP environments. The TNT charge is initiated at the center of the charge. The numerical analyses include eight different L/D ratios ranging from L/D = 1 to L/D = 8. In order to investigate the distribution of blast loads from the CCs in arbitrary orientations, 19 gauge lines are placed at different azimuth angles (θ), ranging from $\theta = 0^\circ$ to $\theta = 90^\circ$ in 5° increments, as shown in Fig. 1. The blast loads along each azimuth angle are recorded within the scaled distance from Z = 0.3 to Z = 10 m/kg$^{1/3}$, at intervals of 0.1 m/kg$^{1/3}$. For comparison purposes, a 1 kg TNT SC with a center-initiated is adopted as the baseline.

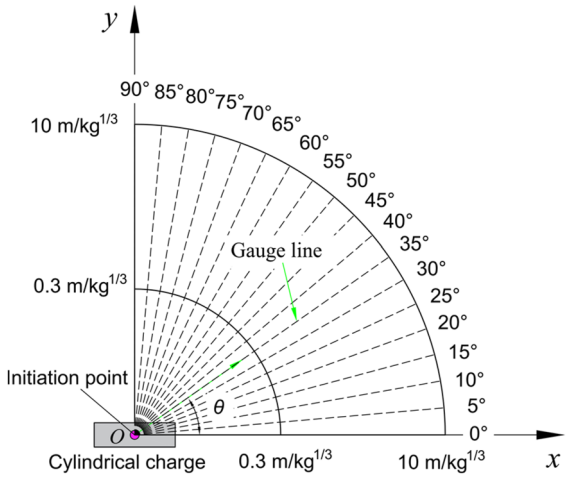


FIG. 1. Azimuth angles (gauge lines) of CCs.

The remapping strategy of the AUTODYN software, which takes into account computational cost and calculation accuracy, is utilized to perform numerical simulations of CCs blast wave propagation across both near-field and far-field regions.[17] In this strategy, two 2D axisymmetric finite element models are set up, as shown in Figs. 2(a) and 2(b). In these figures, the plane x = 0 represents the axis of mirror symmetry, while the plane y = 0 is the axis of the symmetric boundary. Flow-out boundary conditions are applied to the remaining boundary planes of the computational air region. As illustrated in Fig. 2(c), the remapping strategy is executed in two phases: (1) In phase I, the initial product evolution is performed until the blast wave reaches the boundary, with the air region having dimensions of 500 × 500 mm^2. (2) In phase II, the outcome of phase I is remapped into a domain composed exclusively of air, with dimensions of 10 000 × 10 000 mm^2, allowing for the further propagation of the blast wave. Research findings suggest that mesh sizes of 0.5 and 10 mm should be employed for the phase I and phase II modeling, respectively.[11]

B. Material models

The numerical model incorporates two materials: TNT explosive and air. The TNT explosive is modeled employing the Jones–Wilkins–Lee (JWL) equation of state, with the corresponding parameters sourced from the AUTODYN material model library.[18]

The air under different LTP conditions is modeled using the ideal gas equation of state as follows:[18]

$$p_a = (\gamma - 1)\rho_a e, \tag{1}$$

where γ and ρ_a are the adiabatic exponent and density of the air, respectively. Meanwhile, e represents the internal energy per unit mass of the air, defined as[19]

$$e = c_V T_a, \tag{2}$$

where T_a denotes the ambient temperature and c_V is the constant-volume specific heat of air, which varies with ambient temperature T_a.

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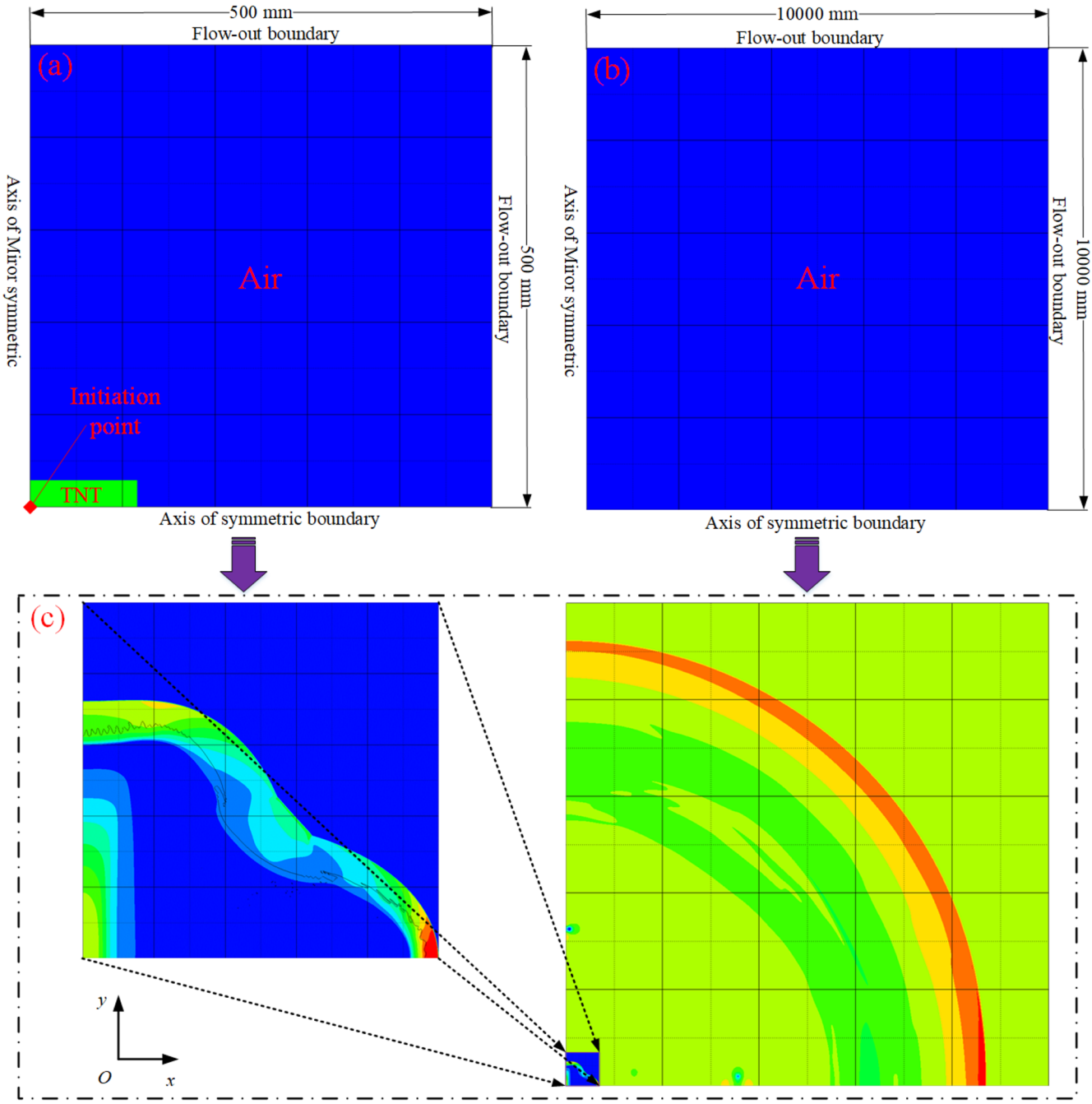


FIG. 2. AUTODYN numerical model: (a) phase I, (b) phase II, and (c) schematic representation of remapping.

The high-altitude environment features a distinctive atmospheric condition characterized by the combined effects of reduced temperature and diminished pressure. The air temperature T_h at different altitudes h is related as[20]

$$T_h = 288.15 - 0.0065h. \tag{3}$$

The air pressure p_h at a specific altitude h is related by the following equation:[20]

$$p_h = 101.32\left(1 - 0.0065\frac{h}{288.15}\right)^{5.2561}, 0m \le h \le 11000m. \tag{4}$$

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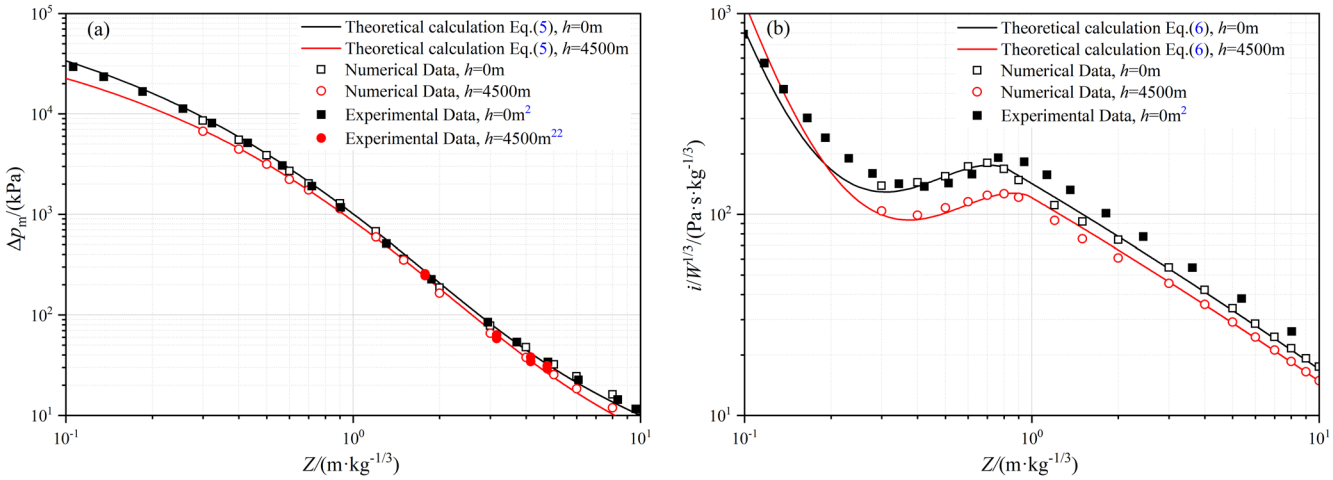


FIG. 3. Theoretical, numerical, and experimental comparisons of SC blast parameters: (left) peak overpressure and (right) impulse.

C. Model validation

Li *et al.*[21] developed equations to predict the peak overpressure (Δp_m) and specific impulse (i) generated by SC in high-altitude environments, i.e.,

$$\frac{\Delta p_m}{p_h} = \frac{808\left[1+\left(\frac{Z}{4.5}\right)^2\left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}\right]}{\sqrt{1+\left(\frac{Z}{0.048}\right)^2\left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}}\sqrt{1+\left(\frac{Z}{0.32}\right)^2\left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}}\sqrt{1+\left(\frac{Z}{1.35}\right)^2\left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}}};\ 0.053\left(\frac{p_h}{p_0}\right)^{-\frac{1}{3}} \le Z \le 500\left(\frac{p_h}{p_0}\right)^{-\frac{1}{3}}, \tag{5}$$

$$\frac{i}{W^{1/3}} = \begin{cases} \left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}\left(\frac{T_0}{T_h}\right)^{\frac{1}{2}} 10^{-3.423\left\{\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]\right\}^4-10.143\left\{\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]\right\}^3-7.558\left\{\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]\right\}^2-1.614\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]+2.14}; Z < 0.8\left(\frac{p_h}{p_0}\right)^{-\frac{1}{3}}, \\ \left(\frac{p_h}{p_0}\right)^{\frac{2}{3}}\left(\frac{T_0}{T_h}\right)^{\frac{1}{2}} 10^{-0.07\left\{\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]\right\}^2-0.853\lg\left[Z\left(\frac{p_h}{p_0}\right)^{\frac{1}{3}}\right]+2.153}; Z \ge 0.8\left(\frac{p_h}{p_0}\right)^{-\frac{1}{3}}, \end{cases} \tag{6}$$

where T_h, T_0 and p_h, p_0 represent the ambient temperatures and pressures under high-altitude and NTP conditions, respectively. W and $Z = r/W^{1/3}$ represent the charge weight and scaled distance, respectively.

Calculate the atmospheric pressure p_h and ambient temperature T_h at high-altitudes of 0 and 4500 m by utilizing Eqs. (3) and (4). Substitute these values into Eqs. (5) and (6) to obtain the SC blast wave overpressure and impulse at various altitudes. Numerical simulations were conducted to examine the SC explosion at various high-altitudes using the AUTODYN software. The blast wave parameters obtained from the simulations, as depicted in Fig. 3, were compared with theoretical calculations and experimental data.[2,22] It is seen that the numerical results for the overpressures and impulse at different high-altitudes agree well with both the theoretical and the experimental results. This agreement confirms that the choice of 0.5 and 10 mm mesh sizes is deemed suitable for investigating blast wave propagation during Phases I and II, respectively.

The analysis of the results presented in Fig. 3 demonstrates a clear trend: as the altitude increases, the overpressure and impulse of the blast wave decrease. This observation indicates that high-altitude environments have a substantial effect on the blast loads.

III. THEORETICAL ANALYSIS OF BLAST PARAMETERS

A. Dimensional analysis of blast parameters

The blast parameters (peak overpressure Δp_m and impulse i) generated from a center-initiated CC in free air depend on the explosive total energy E, the azimuth angle θ, the aspect ratio L/D, the air pressure p_a, the air density ρ_a, and the blast wave propagation distance r, i.e.,

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TABLE I. Dimensionless π-groups of the eight parameters.

	E	p_a	ρ_a	r	L/D	θ	Δp_m	i
M	1	1	1	0	0	0	1	1
T	−2	−2	0	0	0	0	−2	−1
L	2	−1	−3	1	0	0	−1	−1

TABLE II. Dimensionless π-groups of the eight parameters (elementary transformation).

	E	p_a	ρ_a	r	L/D	θ	Δp_m	i
E	1	0	0	1/3	0	0	0	1/3
p_a	0	1	0	−1/3	0	0	1	1/6
ρ_a	0	0	1	0	0	0	0	1/2

$$(\Delta p_m, i) = f(E, \theta, L/D, p_a, \rho_a, r). \tag{7}$$

The eight physical parameters in Eq. (7) are associated with three fundamental dimensions: M, T, and L, which correspond to three independent reference physical parameters. In this case, explosive energy E, air pressure p_a, and air density ρ_a are selected as the fundamental reference parameters, with the corresponding dimensionless π-groups for the eight parameters outlined in Table I.

By applying elementary matrix transformations to Table I, we obtain Table II.

Based on the π theorem,[23] the dimensionless expressions for peak overpressure (Δp_m) and impulse (i) can be obtained from Table II as follows:

$$\left(\frac{\Delta p_m}{p_a}, \frac{i}{E^{1/3} p_a^{1/6} \rho_a^{1/2}}\right) = f\left(\frac{r}{E^{1/3}} p_a^{1/3}, \frac{L}{D}, \theta\right). \tag{8}$$

For air, the EOS $p_a = \rho_a R T_a$ can be used for its description. The energy E released by a given explosive is primarily determined by the charge mass W. Consequently, Eq. (8) can be transformed into

$$\left(\frac{\Delta p_m}{p_a}, \frac{T_a^{1/2} i}{W^{1/3} p_a^{2/3}}\right) = f\left(Z \cdot p_a^{1/3}, \frac{L}{D}, \theta\right), \tag{9}$$

where Z is the scaled distance, $Z = r/W^{1/3}$.

According to the empirical formula for blast parameters in Ref. 11, the scaled distance Z is independent of the quantities L/D and θ. Therefore, Eq. (9) can be expressed as

$$\left(\frac{\Delta p_m}{p_a}, \frac{T_a^{1/2} i}{W^{1/3} p_a^{2/3}}\right) = f\left(Z \cdot p_a^{1/3}\right) \cdot g\left(\frac{L}{D}, \theta\right). \tag{10}$$

B. Blast parameters in LTP environments

Gao *et al.*[11] conducted a comprehensive investigation into the peak overpressure of blast waves generated by center-initiated CCs with $L/D \geq 2$ in NTP environments and developed a calculation formula, as follows:

$$\Delta p_m = \sum_{i=1}^{3} \left[\left(a_{i,1} + a_{i,2} \cdot \ln(L/D) + \sum_{j=1}^{4} \left(a_{i,j,1} + a_{i,j,2} \cdot \ln(L/D) \right) \cdot \sin(j\theta) \right) / Z^i \right], \tag{11}$$

where θ is the azimuth angle, extending from 0° to 90°. The curve-fitting constants $a_{i,1}$, $a_{i,2}$, $a_{i,j,1}$, and $a_{i,j,2}$ are presented in Table 2 in Ref. 11. Equation (11) is a piecewise function that takes into account scaled distances Z_1, Z_2, and Z_p, expressed as follows:[11]

$$\begin{cases} Z_1 = 0.613 + 5.411 e^{-0.962(L/D)}, \\ Z_2 = 10.116 + 49.5 e^{-0.341(L/D)} - 51.942 e^{-0.235(L/D)}, \\ Z_p = 4.749 + 22.033 e^{-0.758(L/D)}. \end{cases} \tag{12}$$

The scaled distance Z for Eq. (12) is divided into three ranges: 0.3 m/kg$^{1/3}$ $\leq Z \leq Z_1$, $Z_1 \leq Z \leq \min(Z_2, Z_p)$, and $Z_2 \leq Z \leq Z_p$ if $Z_2 \leq Z_p$.

Gao *et al.*[11] also derived an empirical formula to calculate the impulse of blast waves from CCs ($L/D \geq 2$) in NTP environments, as

$$i = \sum_{i=1}^{3} \left[\left(a_{i,1} + a_{i,2} \cdot \ln(L/D) + \sum_{j=1}^{4} \left(a_{i,j,1} + a_{i,j,2} \cdot \ln(L/D) \right) \cdot \sin(j\theta) \right) / Z'^i \right], \tag{13}$$

where the azimuth angle θ ranges from 25° to 90°. $Z' = r/W^{2/3}$ is a scaled factor. The constants $a_{i,1}$, $a_{i,2}$, $a_{i,j,1}$, and $a_{i,j,2}$ for different i and j are shown in Table 3 in Ref. 11.

Equations (11) and (13) are only applicable for describing the overpressure and impulse from center-initiated CCs in NTP environments and do not account for the changes in LTP environments. Therefore, based on the dimensional analysis in Eq. (10), Eq. (11) can be reorganized to obtain the analytical expression for the overpressure Δp_m of CCs in LTP environments,

$$\Delta p_m = \sum_{i=1}^{3} \left[\left(a_{i,1} + a_{i,2} \cdot \ln(L/D) + \sum_{j=1}^{4} \left(a_{i,j,1} + a_{i,j,2} \cdot \ln(L/D) \right) \cdot \sin(j\theta) \right) \cdot \left(\frac{p_h}{p_0}\right)^{(1-i/3)} / Z^i \right], \tag{14}$$

where p_h and p_0 represent the ambient pressures under the LTP and NTP conditions, respectively. The three scaled distance ranges Z for Eq. (14) are redefined as follows: $0.3 \cdot (p_0/p_h)^{1/3} \leq Z \leq Z_1 \cdot (p_0/p_h)^{1/3}$, $Z_1 \cdot (p_0/p_h)^{1/3} \leq Z \leq \min(Z_2, Z_p) \cdot (p_0/p_h)^{1/3}$, and $Z_2 \cdot (p_0/p_h)^{1/3} \leq Z \leq Z_p \cdot (p_0/p_h)^{1/3}$ if $Z_2 \leq Z_p$.

Similarly, by performing dimensional analysis on Eq. (10), Eq. (13) can be rearranged to derive the calculation formula for the impulse i of center-initiated CC in different LTP environments,

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$$i = \sum_{i=1}^{3} \left[\left(a_{i,1} + a_{i,2} \cdot \ln(L/D) + \sum_{j=1}^{4} \left(a_{i,j,1} + a_{i,j,2} \cdot \ln(L/D) \right) \cdot \sin(j\theta) \right) \cdot \left(\frac{T_0}{T_h} \right)^{1/2} \cdot \left(\frac{p_0}{p_h} \right)^{(i-1)/3} / Z^i \right], \quad (15)$$

where T_h, T_0 and p_h, p_0 represent the ambient temperatures and pressures under LTP and NTP conditions, respectively.

IV. RESULTS AND DISCUSSION

A. Blast parameters in high-altitude environments

1. Blast wave pressure contours and action ranges

This section examines the characteristics of blast waves from center-initiated CCs at altitudes of 0, 5000, and 10 000 m employing the AUTODYN software. The pressure contours of CCs with L/D = 2, 4, 6, and 8 at different times for different high-altitudes

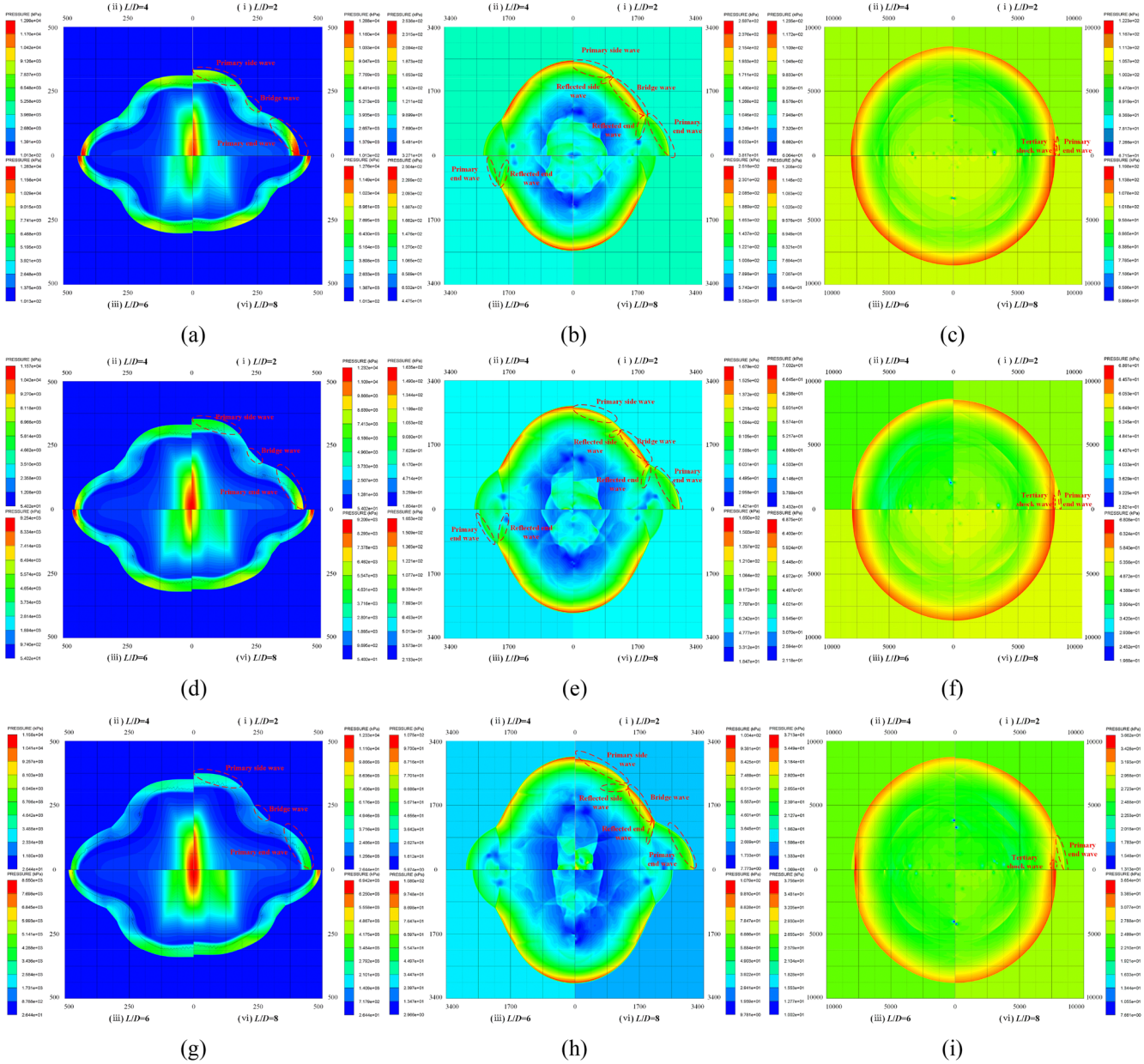


FIG. 4. Pressure contours for L/D = 2, 4, 6, and 8 at different high-altitudes. (a) h = 0 m; t = 0.08 ms, (b) h = 0 m; t = 2.5 ms, (c) h = 0 m; t = 18 ms, (d) h = 5000 m; t = 0.08 ms, (e) h = 5000 m; t = 2.5 ms, (f) h = 5000 m; t = 18 ms, (g) h = 10 000 m; t = 0.08 ms, (h) h = 10 000 m; t = 2.5 ms, and (i) h = 10 000 m; t = 18 ms.

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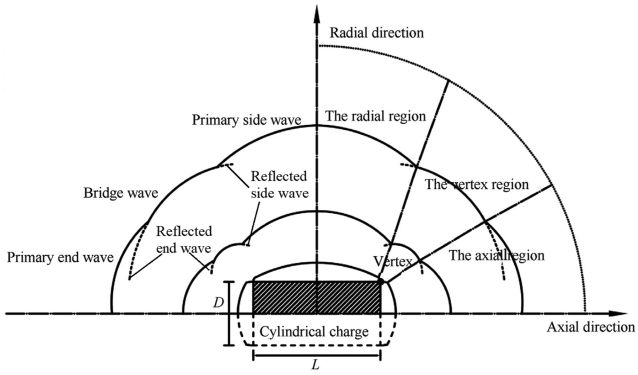


FIG. 5. Development of blast waves and regionalization.[11,24]

are shown in Fig. 4. The blast waves, observed from Fig. 4, generated by center-initiated CC do not propagate into the air as spherical waves, nor do they emerge simultaneously across the entire surface of the charge. It becomes evident that there are three distinct shock waves, labeled as the primary side wave, bridge wave, and primary end wave, depicted in Figs. 4(a-i), 4(d-i), and 4(g-i). These three discernible waves present a "multi-leafed rose" shape pressure contour, which has been experimentally observed in Ref. 24, generated in three distinct regions around the CCs: the radial region, vertex region, and axial region, as depicted in Fig. 5.

As the blast wave propagates, as illustrated by the dashed lines in Fig. 5, the bridge wave along the axial region interacts with the primary end wave, leading to the generation of a reflected end wave. Likewise, the reflection from the other portion of the bridge wave positioned along the radial region and primary side wave creates a reflected side wave [see Figs. 4(b-i), 4(e-i), and 4(h-i)]. The bridge wave positioned along the radial region closely advances alongside the primary side wave. This causes the reflected side wave to gradually fade away as both waves propagate [see Figs. 4(b-iii), 4(e-iii), and 4(h-iii)]. However, as the reflected end wave continues its propagation in the axial direction, the reflected end waves on either side of the symmetry axis converge, resulting in a noticeable enhancement. This enhanced blast then propagates forward, ultimately giving rise to the tertiary shock wave [see Figs. 4(c-i), 4(f-i), and 4(i-i)]. The tertiary shock wave travels at a swifter pace and will ultimately overtake the primary end wave, eventually transforming into a spherical wave [see Figs. 4(c-ii)–4(c-iv)]. Therefore, the blast wave generated by the CCs transforms from its initial "multi-leafed rose" shape into a spherical blast wave over time.

The blast wave action ranges of the different L/D ratio CCs (SC as the baseline) at different high-altitudes for the three different propagation times in Fig. 4 are computed and depicted in Fig. 6. It can be seen that, based on Figs. 4 and 6, as altitude increases for the same propagation time, the blast action range increases, but the overpressure decreases (seen in Fig. 4). The average increase in blast wave action ranges, as the altitude ascends from 0 to 10 000 m, amounts to 39.5%, 57.0%, and 6.0% for the three propagation times, respectively. The average differences of blast wave range between CCs and SC at the three times are 9.8%, 1.2%, and 0.1% for h = 0 m; 10.3%, 1.3%, and 0.4% for h = 5000 m; and 10.7%, 2.1%, and 0.9% for h = 10 000 m, respectively. As the blast wave advances, the difference in the action range of the blast waves generated by the CC gradually decreases. This suggests that the blast waves produced by the CCs gradually evolve into spherical waves. Within the same propagation time, the disparity of the blast wave action ranges between the CC and SC at highland is greater than at lowland. This indicates that in high-altitude environments, the blast wave generated by the CCs takes longer propagation time and distance to transition into spherical waves.

2. Blast parameters

In order to verify the accuracy of Eq. (14) in estimating the peak overpressure from center-initiated CCs in high-altitude environments, numerical simulations and theoretical calculations were conducted at altitudes of 0, 5000, and 10 000 m. The corresponding comparisons are illustrated in Fig. 7.

Figure 7 shows that the numerical simulations and theoretical calculations of the peak overpressure at different altitudes are in good agreement, validating the predictive capability of Eq. (14) for determining peak overpressure generated from center-initiated

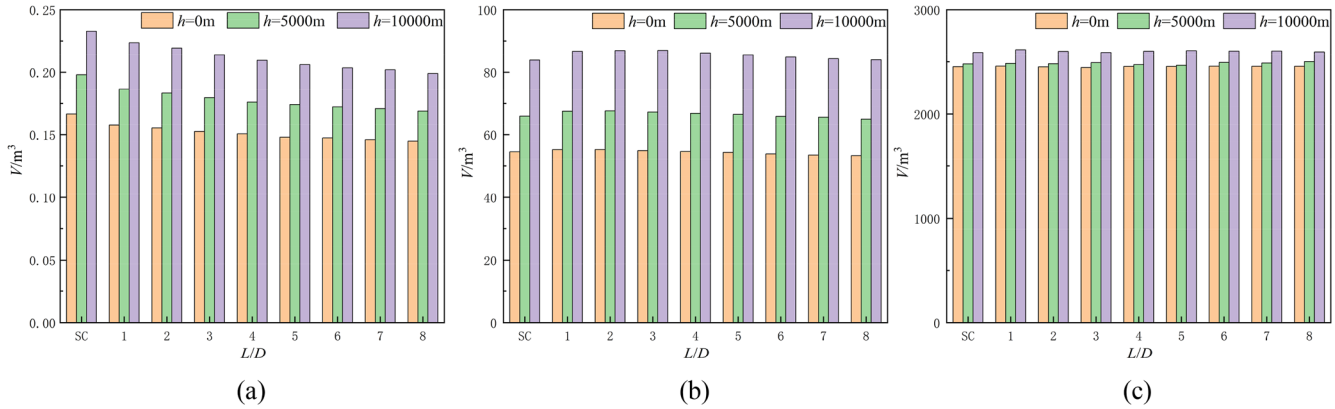


FIG. 6. Action range of blast waves from CCs and SC at different propagation times for various high-altitudes: (a) t = 0.08 ms, (b) t = 2.5 ms, and (c) t = 18 ms.

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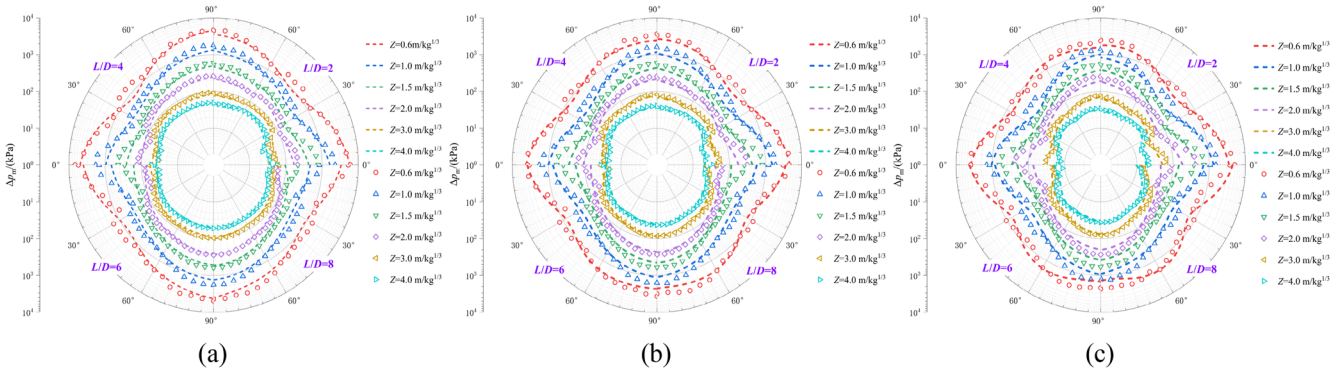


FIG. 7. Comparison of predicted peak overpressures at different high-altitude conditions by Eq. (14) (dashed curves) and numerical data (scatter points): (a) h = 0 m, (b) h = 5000 m, and (c) h = 10 000 m.

CCs under high-altitude environments coupled with low temperature and pressure. Analyzing Fig. 7, it is evident that the overpressure diminishes with altitude. For aspect ratios of L/D = 2, 4, 6, and 8, as the altitude increases from 0 to 10 000 m, the peak overpressure decreases by an average of 28.3%, 29.7%, 29.5%, and 30.8%, respectively. This indicates that the reduced temperature and diminished pressure of the high-altitude environment significantly influence the peak overpressure from CCs. Figure 7 illustrates that at larger scaled distances (e.g., Z = 4.0 m/kg$^{1/3}$), the peak overpressure of CCs with varying aspect ratios becomes nearly uniform in all directions, indicating its transition into a spherical form. Therefore, Eq. (14) can predict the peak overpressure generated by center-initiated CCs ($L/D \geq 2$) in arbitrary orientations at small scaled distances. For large-scaled distances ($Z \geq 4.0$ m/kg$^{1/3}$), the overpressure can be ascertained using the formula for SC with equal mass, such as Eq. (5).

To verify the accuracy of Eq. (15) in predicting the impulse of CCs in high-altitude environments, both simulations and theoretical calculations were also conducted at altitudes of 0, 5000, and 10 000 m. Figure 8 shows that the theoretical and numerical results of the impulse i at different altitudes are in good agreement, indicating that Eq. (15) can effectively predict the impulse from CCs blast waves in high-altitude environments characterized by LTP.

Upon analyzing Fig. 8, it is observed that the impulse decreases with increasing altitude. For aspect ratios of L/D = 2, 4, 6, and 8, the impulse decreases by an average of 39.3%, 40.7%, 40.4%, and 39.7%, respectively, as the altitude changes from 0 to 10 000 m. This indicates that the low-pressure and low-temperature of high-altitudes significantly impact the impulse of the blast wave from CCs. Figure 8 also shows that at larger distances (e.g., Z = 4.0 m/kg$^{1/3}$) for CCs with different L/D ratios, the impulse tends to assume a spherical form. It is important to note that Eq. (15) can be applied to estimate the impulse produced by center-initiated CCs ($L/D \geq 2$) in any orientation ($25^\circ \leq \theta \leq 90^\circ$) at small scaled distances. For large scaled distances, the impulse can be determined using the formula for SCs with the same mass, such as Eq. (6).

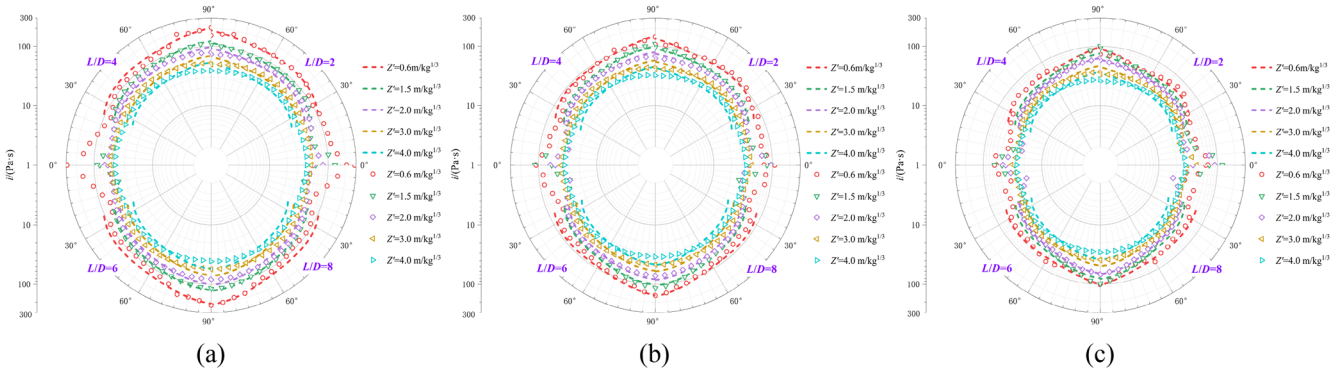


FIG. 8. Comparison of predicted impulses under different high-altitude conditions by Eq. (15) (dashed curves) and numerical data (scatter points): (a) h = 0 m, (b) h = 5000 m, and (c) h = 10 000 m.

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B. Blast wave under LTP conditions decoupling of high-altitudes

1. Blast action ranges

High-altitude environments represent a typical coupled atmospheric environment characterized by reduced temperature and diminished pressure. Both factors significantly affect the blast parameters of CCs, necessitating further analysis to quantify the extent of their influence. To investigate the blast wave characteristics of CCs under low-pressure and low-temperature conditions, atmospheric pressures were selected as 101.325, 54.019, and 26.436 kPa, respectively, with an ambient temperature of 15 °C for low-pressure conditions. Likewise, for low-temperature conditions, the environmental temperatures were set to 15, −17.5, and −50 °C, respectively, maintaining an ambient pressure of 101.325 kPa. These LTP conditions are associated with decoupling at altitudes of 0, 5000, and 10 000 m. The pressure contours for CCs with varying aspect ratios at the same three propagation times (t = 0.08, 2.5, and 18 ms) under different environmental pressures (temperatures) were determined using numerical simulations. Studies have revealed that, akin to the pressure contours of CCs blast waves in high-altitude environments, the pressure contours of CCs with varying L/D ratios under the LTP conditions also display a "multi-leafed rose" pattern with the three distinct waves as shown in Fig. 4. Over time, the "multi-leafed rose" pattern gradually evolves into a spherical shape. The blast wave action range for CCs and SC under varying decoupled environmental temperatures and pressures was statistically analyzed, as illustrated in Fig. 9.

From the blast wave action ranges of CCs with different L/D ratios in Fig. 9, it can be observed that the blast action range decreases with a decrease in ambient temperature, while it increases as environmental pressure decreases. At blast wave propagation times of 0.08, 2.5, and 18 ms, the blast wave action range increases by an average of 50.3%, 86.8%, and 39.6%, respectively, with a reduction in environmental pressure from 101.325 to 26.436 kPa. In contrast, the action range decreases by an average of 7.6%, 18.2%, and 25.9%, respectively, as the environmental temperature drops from 15 to −50 °C.

Based on a comprehensive analysis of Figs. 6 and 9, it is observed that as the altitude increases from 0 to 10 000 m, the blast wave action range extends by an average of 34.2% across the three propagation times. Isolating the effects of low-pressure and low-temperature reveals, as shown in Fig. 9, that a reduction in air pressure (from 101.325 to 26.436 kPa) leads to an average 58.9% increase in the blast wave action range. In contrast, a corresponding decrease in environmental temperature (from 15 to −50 °C)

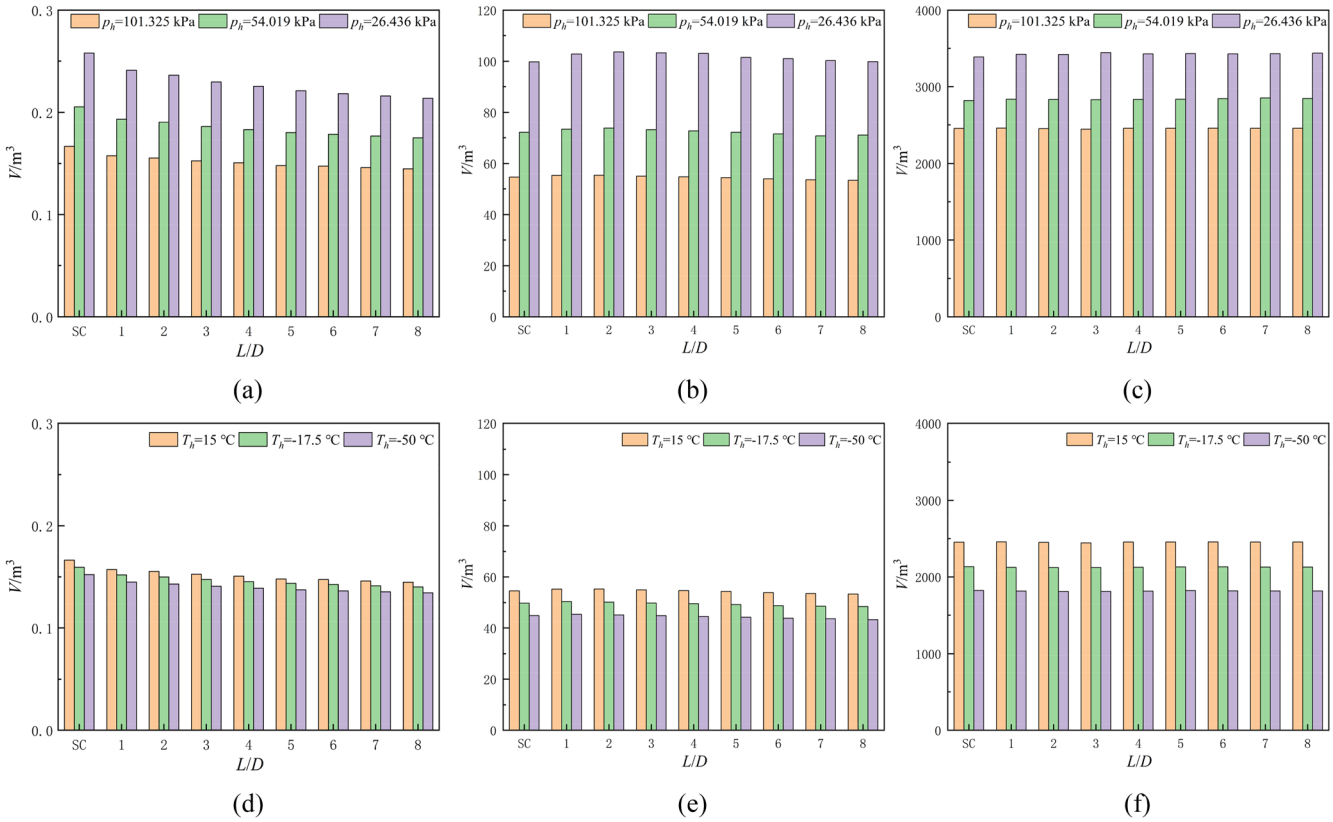


FIG. 9. Action range of blast waves from CCs and SC at different propagation times for various LTPs. (a) t = 0.08 ms; T_h = 15 °C, (b) t = 2.5 ms; T_h = 15 °C, (c) t = 18 ms; T_h = 15 °C, (d) t = 0.08 ms; p_h = 101.325 kPa, (e) t = 2.5 ms; p_h = 101.325 kPa, and (f) t = 18 ms; p_h = 101.325 kPa.

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results in an average 17.2% reduction in the action range. The effect of high-altitude on the blast wave action range exhibits a positive correlation with environmental pressure and a negative correlation with ambient temperature. By comparison, it is evident that in high-altitude environments, environmental pressure exerts a more significant influence on the blast wave's action range than environmental temperature.

2. Blast parameters

To verify the accuracy of Eq. (14) in predicting the peak overpressure from CCs under LTP conditions, theoretical calculations and numerical simulations were conducted. The tests were performed at ambient temperatures of 15, −17.5, and −50 °C (with an ambient pressure of 101.325 kPa), as well as at ambient pressures of 101.325, 54.019, and 26.436 kPa (with an ambient temperature of 15 °C). Figure 10 presents the comparative results of the theoretical and numerical analyses of the CCs peak overpressure (Δp_m) under different LTP conditions. Figure 10 shows the calculations and simulations of the peak overpressure (Δp_m) under varying environmental temperatures and pressures are in close agreement. This indicates that Eq. (14) effectively predicts the peak overpressure from center-initiated CCs in LTP environments. It is worth noting that Eq. (14) can be applied to estimate the peak overpressure produced by centrally initiated CCs with $L/D \geq 2$ in any orientation at small scaled distances. At large scaled distances ($Z \geq 4.0$ m/kg$^{1/3}$), the overpressure becomes nearly uniform in all directions and can be determined using Eq. (5) for SCs with the same mass.

Analysis of Fig. 10 shows that when the aspect ratios L/D = 2, 4, 6, and 8, the average overpressure decreases by 28.5%, 29.9%, 30.7%, and 30.9%, respectively, when the ambient pressure drops from 101.325 to 26.436 kPa. Similarly, as the environmental temperature drops from 15 to −50 °C, the average peak overpressure increases by 0.6%, 0.7%, 0.4%, and 0.5%, respectively. The peak overpressure of CC blast waves shows a slight decrease with a reduction in ambient temperature and a significant decrease as environmental pressure decreases. As shown in Figs. 7 and 10, the peak overpressure decreases by an average of 29.6% as the altitude rises from 0 to 10 000 m. This reduction is primarily due to two factors: 30.0% is primarily caused by the decrease in environmental pressure, while 0.5% is attributed to the drop in ambient temperature. These results highlight that in high-altitude environments, the reduction in environmental pressure plays a more dominant role in influencing the blast overpressure of CCs compared to the decrease in environmental temperature.

Similarly, the theoretical calculations and numerical simulation results of the impulse of CCs blast waves under LTP conditions decoupling of different altitudes were statistics, as shown in Fig. 11. Figure 11 shows that the theoretical and numerical results of the impulse under varying environmental temperatures and pressures are in close agreement. This indicates that Eq. (15) effectively predicts the impulse from CCs in LTP environments. Figure 11 also

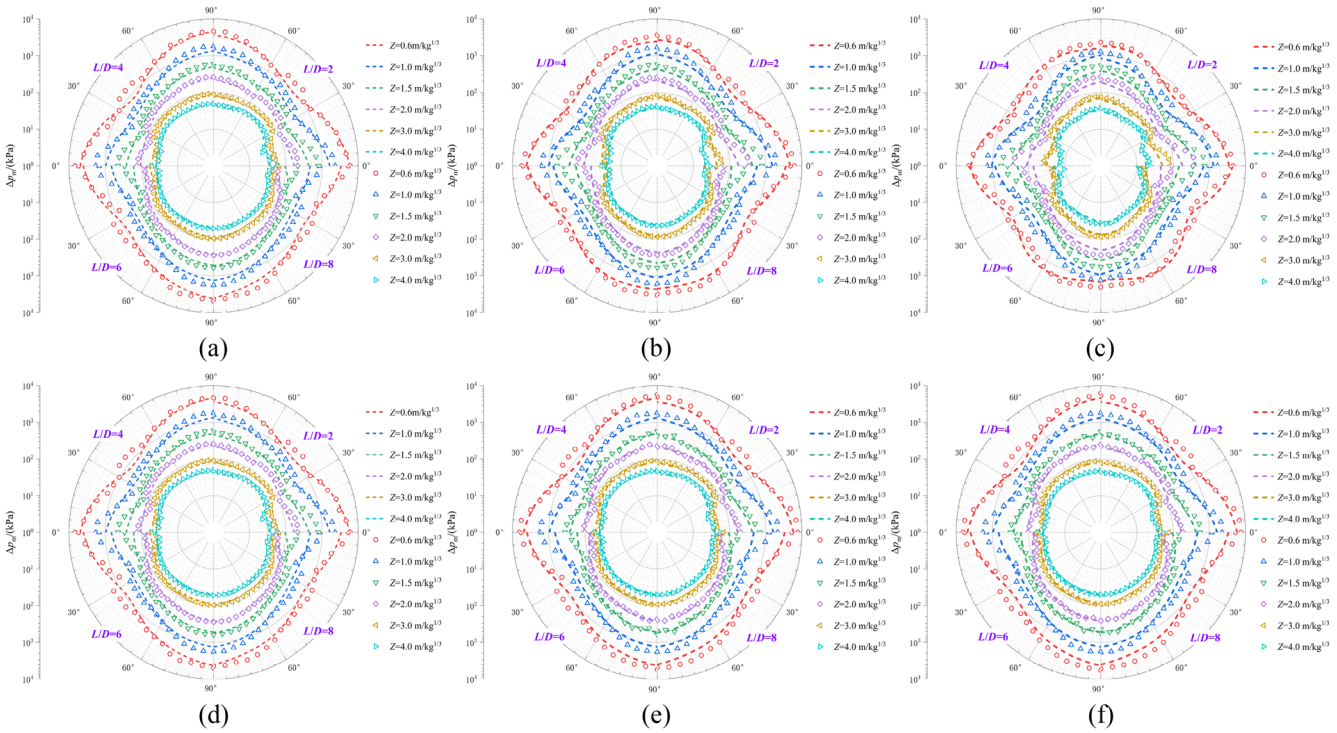


FIG. 10. Comparison of predicted peak overpressures under different LTP conditions by Eq. (14) (dashed curves) and numerical data (scatter points): (a) p_h = 101.325 kPa, (b) p_h = 54.019 kPa, (c) p_h = 26.436 kPa, (d) T_h = 15 °C, (e) T_h = −17.5 °C, and (f) T_h = −50 °C.

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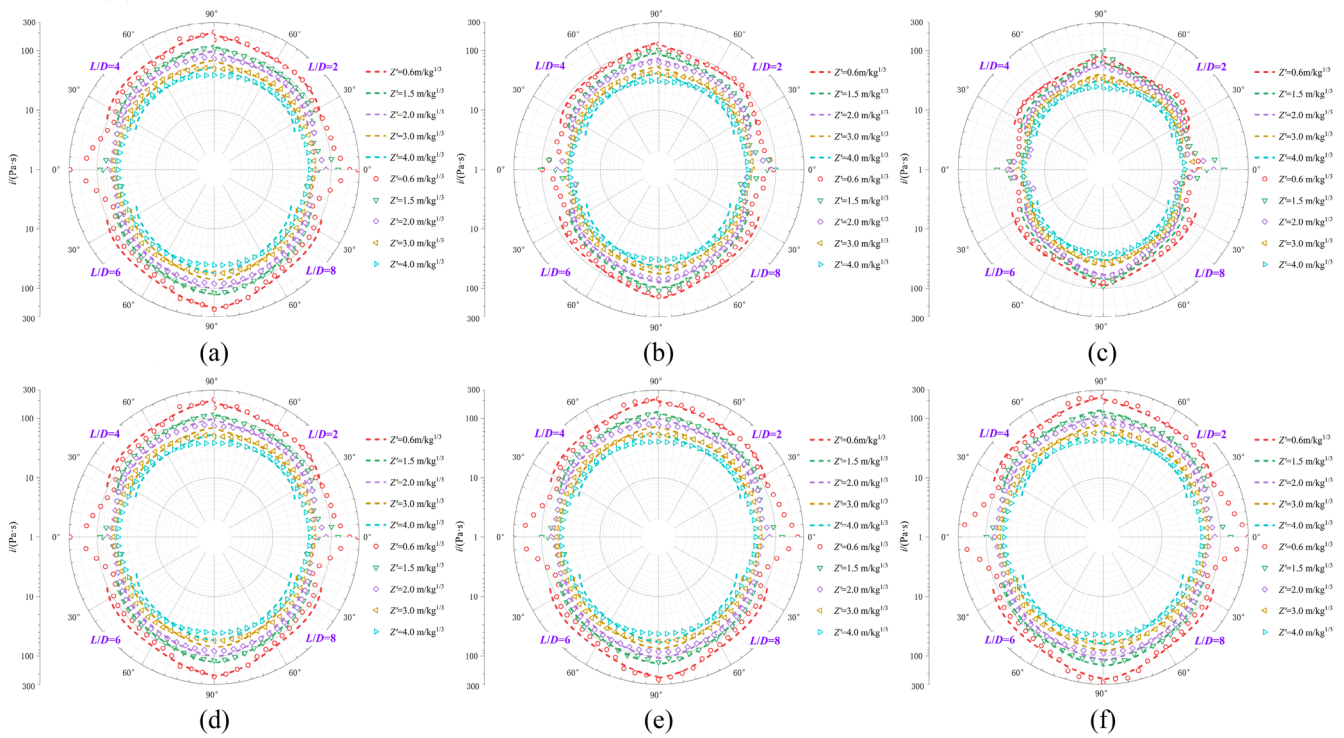


FIG. 11. Comparison of predicted impulses under different LTP conditions by Eq. (15) (dashed curves) and numerical data (scatter points): (a) p_h = 101.325 kPa, (b) p_h = 54.019 kPa, (c) p_h = 26.436 kPa, (d) T_h = 15 °C, (e) T_h = −17.5 °C, and (f) T_h = −50 °C.

illustrates that at larger scaled distances ($Z \geq 4.0$ m/kg$^{1/3}$), the impulse from CCs with varying aspect ratios into a spherical form. Equation (15) serves as a method for predicting the impulse generated from center-initiated CCs ($L/D \geq 2$) in any orientation ($25^\circ \leq \theta \leq 90^\circ$) at small scaled distances. For large scaled distances, the impulse can be determined using Eq. (6) for SCs with the same mass.

Based on the analysis of Fig. 11, when the aspect ratios L/D = 2, 4, 6, and 8, the impulses decrease by an average of 48.4%, 48.6%, 48.2%, and 48.3%, respectively, as the air pressure drops from 101.325 to 26.436 kPa. As the ambient temperature drops from 15 to −50 °C, the impulses increase by an average of 14.1%, 14.6%, 15.7%, and 16.3%, respectively. The impulses of CCs decrease with a reduction in ambient pressure but increase as ambient temperature decreases. Figures 8 and 11 show that as the altitude increases from 0 m to 10,000 m, the impulse decreases by an average of 40.0%. This reduction includes an average of 48.4% decrease due to the drop in pressure, partially offset by an average of 15.2% increase resulting from the decrease in temperature. This indicates that, in high-altitude environments, the reduction in environmental pressure influences the blast impulse of CCs more significantly than the decrease in environmental temperature.

V. CONCLUSIONS

This paper presents analyses of the blast parameters (peak overpressure and specific impulse) and blast action range produced by center-initiated CCs in a high-altitude environment using the AUTODYN software and dimensional analysis theory. The following important conclusions are drawn:

(1) The empirical formula for predicting free air blast parameters in low temperature and low pressure environments, generated by center-initiated CCs with varied aspect ratios in arbitrary orientations, is derived by employing dimensional analysis theory. The predictive reliability of the formula is verified through numerical modeling of CC blast waves at different high-altitudes, decoupling the corresponding low-temperature and low-pressure conditions.

(2) The blast wave action range produced by CCs expands as the altitude increases. It also increases with decreasing environmental pressure but diminishes with lower environmental temperatures. The high-altitude increased from 0 to 10 000 m, and the blast wave action range increased by 34.2% on average. Among them, the average increase in ambient pressure is 58.9%, while the average reduction in environmental temperature is 17.2%. The impact of low-pressure conditions on the blast wave action range in high-altitude environments is higher than that in low-temperature conditions.

(3) The peak overpressure (impulses) of the blast wave generated by the CC decreased by an average of 29.6% (40.0%) for altitude increases from 0 to 10 000 m. Specifically, the ambient pressure decoupled from the altitude's effect decreased

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by an average of 30.0% (48.4%), while the ambient temperature decoupled from the altitude led to an increase of 0.5% (15.2%). These results highlight that in high-altitude environments, the reduction in environmental pressure plays a more dominant role in influencing the blast parameters of CCs compared to the decrease in environmental temperature.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Rui Li: Funding acquisition (equal); Methodology (equal); Writing – original draft (equal). **Rui Sun**: Formal analysis (equal); Writing – original draft (supporting). **Ming-hang Hu**: Investigation (equal). **Ya-jing Chen**: Data curation (equal); Investigation (equal). **Quan Wang**: Methodology (equal); Supervision (equal). **Dong-Dong Zhang**: Software (equal); Validation (equal). **Xue-jiao Li**: Funding acquisition (equal). **Xiao-rong Cui**: Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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